VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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A PROJECT WORK PHASE

"Automatic lymphocyte detection on gastric cancer IHC images using deep learning"

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

Submitted by

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1. INTRODUCTION

Gastric cancer is one of the leading causes of cancer-related mortality worldwide. The tumor microenvironment plays a crucial role in disease progression and treatment response. Among various immune cells, **Tumor-Infiltrating Lymphocytes (TILs)** are considered important biomarkers for:

- Patient prognosis
- Immunotherapy response prediction
- Survival rate estimation

Traditionally, pathologists manually examine hematoxylin and eosin (H&E) stained histopathology slides under a microscope to identify and quantify lymphocytes. However, manual assessment is:

- Time-consuming
- Subjective
- Prone to inter-observer variability

With the advancement of digital pathology and deep learning, automatic lymphocyte detection systems can assist pathologists by providing accurate, fast, and reproducible analysis.

In today's digital era, organizations and individuals generate and consume vast amounts of information in the form of documents. However, traditional document management systems treat documents as static files, limiting users to basic reading and manual analysis. Extracting meaningful insights from large collections of PDFs, reports, research papers, and text files can be time-consuming and inefficient.

2. MOTIVATION BEHIND THE PROJECT WORK

The motivation for this project arises from the following clinical and technical challenges:

🔬 Clinical Motivation

- Early and accurate immune profiling improves treatment planning.

- Quantifying TILs helps predict response to immunotherapy.
- Manual evaluation is inconsistent across observers.

💻 **Technical Motivation**

- Whole Slide Images (WSIs) are extremely large (gigapixel scale).
- Lymphocytes are very small and densely packed.
- Variability in staining affects detection accuracy.
- Recent advances in AI (CNNs, U-Net, YOLO, Transformers) show high potential in medical image analysis.

Thus, an automated system can:

- Reduce pathologist workload
- Improve reproducibility
- Provide objective immune scoring
- Enable large-scale clinical studies

3. PROBLEM STATEMENT DEFINITION

Manual identification and quantification of lymphocytes in gastric cancer histopathology slides is labor-intensive, subjective, and not scalable for large clinical datasets.

Therefore, the problem can be defined as:

To design and develop an automated deep learning-based system that accurately detects and quantifies lymphocytes in H&E-stained gastric cancer histopathology images.

The system must:

- Detect small lymphocyte nuclei
- Differentiate lymphocytes from tumor and stromal cells
- Handle stain variability

4. STATEMENT OF OBJECTIVES

The main objectives of this project are:

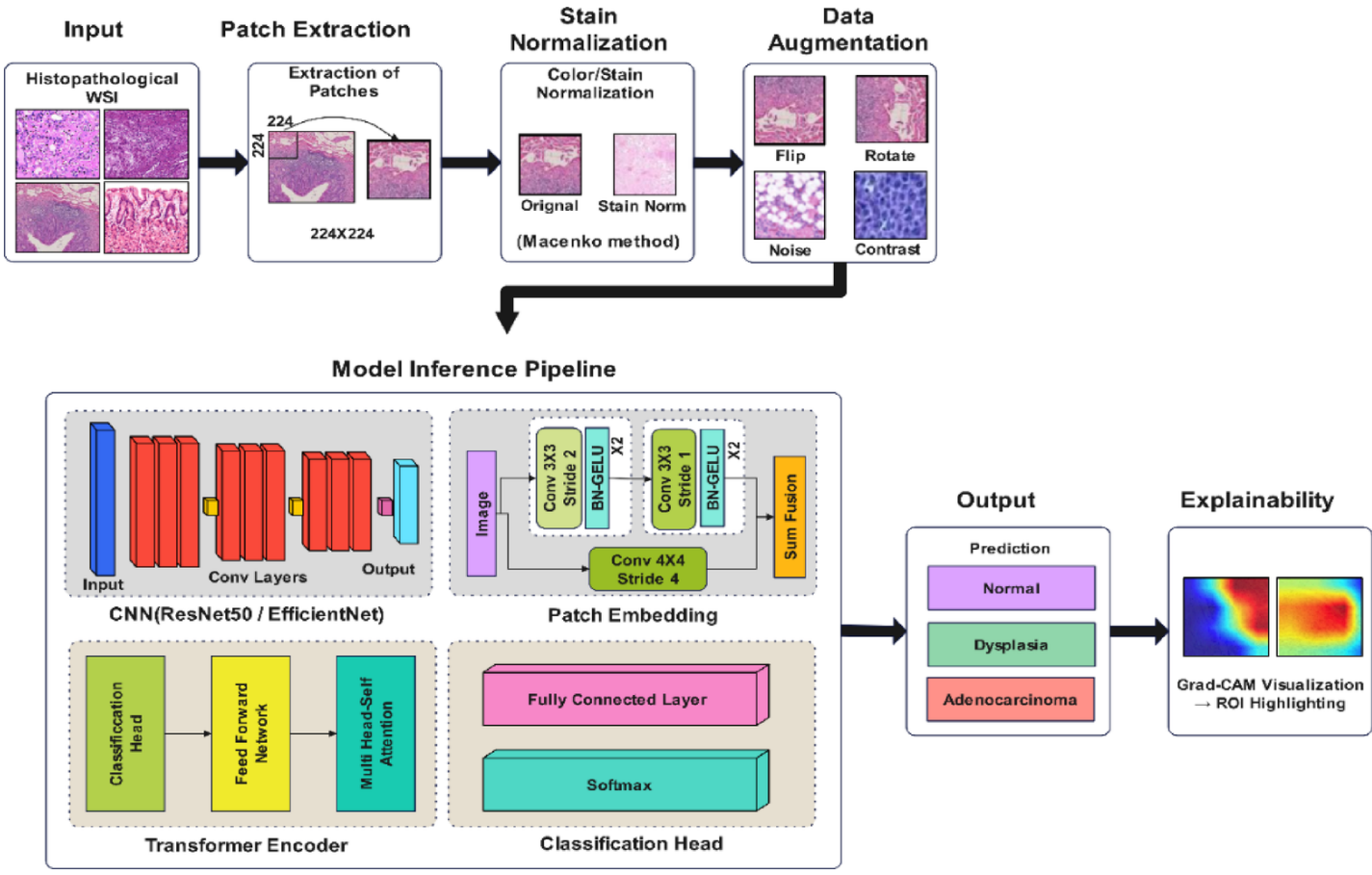
Primary Objectives

1. To develop a preprocessing pipeline for histopathology images.
2. To design and train a deep learning model for lymphocyte detection.
3. To accurately classify lymphocytes from other cell types.
4. To evaluate model performance using standard metrics.

Secondary Objectives

5. To quantify lymphocyte density per tumor region.
6. To reduce inter-observer variability.
7. To optimize the system for computational efficiency.
8. To build a visualization interface for detected lymphocytes.

5. SYSTEM ARCHITECTURE



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